Nucleos activos de Galaxias

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Coeficientes para colisiones

Para el caso de un átomo simple que se encuentra excitado en un nivel k. Al realizar una transición de k a J, se produce una emisión espontánea de un fotón. La energía por unidad de tiempo, por unidad de volumen y por unidad de ángulo sólido, está dada por:

$$j_\nu = N_{k} A_{kj} \frac{h\nu_{kj}}{4\pi} \qquad (1)$$

N_{k}: Número de partículas en el estado k.

A_{kj}: Coef de Einstein para un decaimiento espontáneo

$h\nu_{kj}$: Energía emitida por el fotón con frecuencia ν_{kj}.

Las transiciones hacia arriba $j \to k$ deben ser iguales a las trancisiones hacia abajo $k \to j$

↳ Poblar

↳ despoblar

Para un nivel excitado $\uparrow_{jk}$. El decaimiento puede darse por:

- Nueva colisión $\downarrow_{kj}$
- Emisión expontánea $\rightsquigarrow_{kj}$

$$\hookrightarrow \quad \uparrow_{jk} = \downarrow_{kj} + \rightsquigarrow_{kj} \qquad (2)$$

Buscamos el coeficiente de exitación inelástica γ_{jk} y el coeficiente para desexitación Superelástica γ_{kj}. Para N_e partículas en un tiempo t (tiempo promedio de cada colisión)

↳ Tasa de colisiones:

$$q = \frac{N_e}{t} = N_i N_e (V_r \sigma) \qquad (3)$$

↳ Sección transversal de Colisión

$$\sigma_{jk} \propto \pi\lambda^2 \propto \pi \left(\frac{\hbar}{mv_r}\right)^2$$

En función de la intensidad de colisión $\Omega(jk)$:

$$\sigma_{jk} = \frac{1}{g_j} \frac{\pi\hbar^2}{m^2 v_{rt}^2} \Omega_{kj}$$

$$\hookrightarrow \quad \gamma_{jk} = \langle V_{rt}\sigma_{jk}\rangle = \frac{\int_0^\infty F(V_{rt}) V_{rt} \sigma_{jk} \, dV_{rt}^3}{\int_0^\infty F(V_{rt}) dV_{rt}^3} = \frac{4L^3}{\sqrt{\pi}} \int_0^\infty V_{rt}^3 \sigma_{jk} e^{-L^2 V_{rt}^2} dV_{rt}^3$$

$$L = \frac{m_r}{2KT} \quad ; \quad L^2 = \frac{m_{pp} m_{pc}}{(m_{pp}+m_{pc})} \frac{1}{2KT}$$

En 3.

$$q_{jk} = N_{it} N_e \langle V_{rt}\sigma_{jk}\rangle$$

Si los estados exitados regresan al estado inicial por colisión:

$$q_{kj} = N_{i\downarrow} N_e \langle V_{r\downarrow}\sigma_{kj}\rangle$$

$$\sigma_{kj} = \frac{1}{g_k}\frac{\pi\hbar^2}{m^2 v_{r\downarrow}^2} \Omega_{jk}$$

$$\Omega_{jk} = \Omega_{kj}$$

Si decaen de manera espontánea por la desexitación de un fotón:

$$q_{\vec{kj}} = N_{i\downarrow} A_{kj}$$

La ecuación de equilibrio estadístico (2) queda:

$$q_{jk} = q_{kj} + q_{\vec{kj}}$$

$$N_{i\uparrow} N_e \langle V_{r\uparrow} \sigma_{jk} \rangle = N_{i\downarrow}\left[N_e \langle V_{r\downarrow} \sigma_{kj} \rangle + A_{kj} \right]$$

$$N_{i\downarrow} = \frac{N_{i\uparrow} N_e \langle V_{r\uparrow} \sigma_{jk} \rangle}{N_e \langle V_{r\downarrow} \sigma_{kj} \rangle + A_{kj}}$$

Entonces, la eq 3 queda:

$$j_\nu = \frac{N_{i\uparrow} N_e \langle V_{r\uparrow} \sigma_{jk} \rangle}{N_e \langle V_{r\downarrow} \sigma_{kj} \rangle + A_{kj}} A_{kj} \frac{h\nu_{kj}}{4\pi} \qquad (4)$$

Casos límites

a) Densidad muy baja

$$J_\nu = N_{i\uparrow} N_e \langle V_{r\uparrow} \sigma_{jk} \rangle \frac{h\nu_{kj}}{4\pi}$$

b) Densidad muy alta

$$J_\nu = N_{i\uparrow} A_{kj} \frac{\langle V_{r\uparrow} \sigma_{jk} \rangle}{\langle V_{r\downarrow} \sigma_{kj} \rangle} \frac{h\nu_{kj}}{4\pi}$$

c) Densidad media (crítica)

$$(N_e)_c = \frac{A_{kj}}{\langle V_{r\downarrow} \sigma_{kj} \rangle}$$

Intensidades de Colisión:

Si $L=0$ ó $S=0$ Las intensidades de colisión pueden calcularse como:

$$\Omega(SLJ, S'L', J') = \frac{(2J'+1)}{(2S'+1)(2L'+1)} \Omega(SL, S'L')$$

Notación espectroscópica

$^{2S+1}L_J$

Por ejemplo, $\Omega(^4S_{3/2}, {}^2D_{5/2})$ y $\Omega(^4S_{3/2}, {}^2D_{3/2})$ teniendo en cuenta que $\Omega(^4S, {}^2d) = 6.9$ y $\Omega(^2d_{3/2}, {}^2d_{5/2}) = 7.47$ y:

$^{2S+1}L_J$:

$L=0$	$S=3/2$	$J=3/2$
$L=2$	$S=1/2$	$J=3/2$
$L=2$	$S=1/2$	$J=5/2$

- $\Omega(^4S_{3/2}, {}^2D_{5/2}) = \Omega(3/2, 0, 3/2)(1/2, 2, 5/2) = \frac{2(5/2+1)}{(2(1/2)+1)(2(2)+1)} \Omega(3/2, 0)(1/2, 2) = 4.14$

$$\Omega(^4S_{3/2},\, ^2D_{3/2}) = \Omega(3/2, 0, 3/2)(1/2, 2, 3/2) = \frac{(2(3/2)+1)}{(2(1/2)+1)(2(2)+1)}\, \Omega(3/2, 0)(1/2, 2) = 2.76$$

Tasas de colisión para valores medios de intensidad:

$$q_{kj} = N_{i\downarrow} N_e \int_0^\infty \frac{4}{\pi^{1/2}} \left(\frac{m}{2KT}\right)^{3/2} V_{r\downarrow}^3\, \sigma_{kj}\, e^{-mV_{r\downarrow}^2/2KT}\, dV_{r\downarrow}$$

$$= \frac{4 N_{i\downarrow} N_e \pi^{1/2} \hbar^2}{m^2} \frac{\Omega_{JK}}{g_K} \left(\frac{m}{2KT}\right)^{3/2} \int_0^\infty V_{r\downarrow}\, e^{-mV_{r\downarrow}^2/2KT}\, dV_{r\downarrow}$$

$$= \frac{4 N_{i\downarrow} N_e \pi^{1/2} \hbar^2}{m^2} \frac{\Omega_{JK}}{g_K} \left(\frac{m}{2KT}\right)^{3/2} \left(\frac{KT}{m}\right)$$

$$= \frac{N_{i\downarrow} N_e (2\pi)^{1/2}}{(KT)^{1/2}} \frac{\hbar^2}{m^{3/2}} \frac{\Omega_{JK}}{g_K}$$

En el sistema CGS $\quad \dfrac{\hbar^2}{m^{3/2}} \sqrt{\dfrac{2\pi}{K}} = 8.6 \times 10^{-6}$

$$\hookrightarrow \quad q_{kj} = N_{i\downarrow} N_e \frac{8.6 \times 10^{-6}}{\sqrt{T}} \frac{\Omega_{JK}}{g_K}$$

Buscamos: $\quad q_{jk} = N_{i\uparrow} N_e \langle V_{r\uparrow} \sigma_{jk} \rangle$

Teniendo en cuenta la eq $\quad g_j \gamma_{jk} = g_k \gamma_{kj} e^{-E_{jk}/KT}$

$$q_{jk} = N_{i\uparrow} N_e \langle V_{r\uparrow} \sigma_{jk} \rangle = N_{i\uparrow} N_e \frac{g_k}{g_j} \gamma_{kj} e^{-E_{jk}/KT}$$

$$= \frac{N_{i\uparrow}}{N_{i\downarrow}} \frac{g_k}{g_j} q_{kj} e^{-E_{jk}/KT}$$

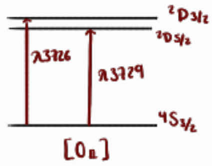
$$q_{kj} = N_{i\uparrow} N_e \frac{8.6 \times 10^{-6}}{T^{1/2}} \frac{\Omega_{JK}}{g_j} e^{-E_{jk}/KT}$$

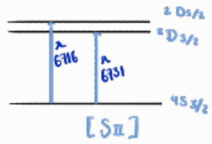
Estudiamos principalmente la **Región de líneas delgadas.** $\quad \hookrightarrow 200 \lesssim \underbrace{\Delta V_{FWHM}}_{\text{Ancho total a la mitad de la altura.}} \lesssim 900\ \mathrm{Km\,s^{-1}}$

$$v = \frac{\Delta\lambda}{\lambda} c \qquad \frac{\Delta\lambda}{\lambda} = \frac{v}{c}$$

$\boxed{H_\alpha = 6583\ \text{Å}}$

Estudiaremos las líneas de emisión de SII y OII:





Para densidades muy bajas:

$$J_\nu = N_{i\uparrow} N_e \langle V_{r\uparrow} \sigma_{jk} \rangle \frac{h\nu_{kj}}{4\pi}$$

Como $g = 2J+1$

SII:
$$\frac{F_{6716}}{F_{6731}} = \frac{(g_K)_{6716}}{(g_K)_{6731}} = \frac{2\left(\frac{5}{2}\right)+1}{2\left(\frac{3}{2}\right)+1} = 1.5$$

Ecuación de balance energético

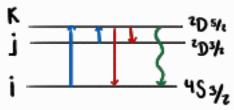
Teniendo en cuenta:

$\vec{q}_{kj} = N_{i\downarrow} A_{kj}$; $q_{kj} = \frac{N_{i\downarrow} N_e}{T^{1/2}} \frac{8.6\times10^{-6}\,\Omega_{jk}}{g_k}$; $q_{jk} = N_{i\uparrow} N_e \frac{8.6\times10^{-6}\,\Omega_{jk}}{T^{1/2}\; g_j} e^{-E_{jk}/KT}$

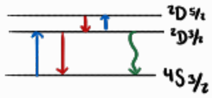
y definiendo $C = \frac{8.63\times10^{-6}}{T^{1/2}}$

λ= 6716
Surge un fotón.

- $6716 \Rightarrow q(^4S_{3/2} \to {}^2D_{5/2}) + q(^2D_{3/2} \to {}^2D_{3/2}) = q(^2D_{5/2} \to {}^4S_{3/2}) + q(^2D_{5/2} \to {}^2D_{3/2}) + q(^2D_{5/2} \to {}^4S_{3/2})$



Para 6731



la λ de este gap es muy pequeña, despreciable

λ= 6731

$6731: q(^4S_{3/2} \to {}^2D_{3/2}) + q(^2D_{5/2} \to {}^2D_{3/2}) = q(^2D_{3/2} \to {}^4S_{3/2}) + q(^2D_{3/2} \to {}^2D_{5/2}) + q(^2D_{3/2} \to {}^4S_{3/2})$

Entonces

$6716 \Rightarrow q(^4S_{3/2} \to {}^2D_{5/2}) + q(^2D_{3/2} \to {}^2D_{3/2}) = q(^2D_{5/2} \to {}^4S_{3/2}) + q(^2D_{5/2} \to {}^2D_{3/2}) + q(^2D_{5/2} \to {}^4S_{3/2})$

En general:

$$q(i\to K) + q(j\to K) = q(K\to i) + q(K\to j) + N_K A_{Ki}$$

$$\hookrightarrow \frac{N_i N_e}{T^{1/2}} \frac{8.6\times10^{-6}\,\Omega_{iK}}{g_i} e^{-E_{iK}/KT} + \frac{N_j N_e}{T^{1/2}} \frac{8.6\times10^{-6}\,\Omega_{jK}}{g_j} e^{-E_{jK}/KT}$$

$$= N_K N_e \frac{8.63\times10^{-6}}{\sqrt{T}} \frac{\Omega_{iK}}{g_K} + N_K N_e \frac{8.63\times10^{-6}}{\sqrt{T}} \frac{\Omega_{jK}}{g_K} + N_K A_{Ki}$$

$$\Rightarrow N_i N_e C \frac{\Omega_{iK}}{g_i} e^{-E_{iK}/KT} + N_j N_e C \frac{\Omega_{jK}}{g_j} e^{-E_{jK}/KT} = N_K \left[N_e C \frac{\Omega_{iK} - \Omega_{jK}}{g_K} + A_{Ki} \right]$$

Como $E_{jK} \ll KT \Rightarrow e^{-E_{jK}/KT} \simeq 1$

$$\hookrightarrow N_i N_e C \frac{\Omega_{iK}}{g_i} e^{-E_{iK}/KT} + N_j N_e C \frac{\Omega_{jK}}{g_j} = N_K \left[N_e C \frac{\Omega_{iK} + \Omega_{jK}}{g_K} + A_{Ki} \right]$$

Entonces:

$$\hookrightarrow \frac{C N_e N_{^4S_{3/2}} \Omega(^4S_{3/2}, {}^2D_{5/2})}{g(^4S_{3/2})} e^{-E(^4S_{3/2}, {}^2D_{5/2})/KT} + \frac{C N_e N_{^2D_{3/2}} \Omega(^2D_{3/2}, {}^2D_{5/2})}{g(^2D_{3/2})} e^{-E(^2D_{3/2}, {}^2D_{5/2})/KT} =$$

$$\frac{C N_e N_{^2D_{5/2}} \Omega(^2D_{5/2}, {}^4S_{3/2})}{g(^2D_{5/2})} + \frac{C N_e N_{^2D_{5/2}} \Omega(^2D_{5/2}, {}^2D_{3/2})}{g(^2D_{5/2})} + N_{^2D_{5/2}} A_{6716}$$

$$\hookrightarrow \frac{C N_e N_{^4S_{3/2}} \Omega(^4S_{3/2}, {}^2D_{5/2})}{g(^4S_{3/2})} e^{-E(^4S_{3/2}, {}^2D_{5/2})/KT} + \frac{C N_e N_{^2D_{5/2}} \Omega(^2D_{3/2}, {}^2D_{5/2})}{g(^2D_{5/2})} =$$

$$\frac{C N_e N_{^2D_{5/2}} \Omega(^2D_{3/2}, {}^4S_{3/2})}{g(^2D_{3/2})} + \frac{C N_e N_{^2D_{3/2}} \Omega(^2D_{3/2}, {}^2D_{5/2})}{g(^2D_{3/2})} e^{-E(^2D_{3/2}, {}^2D_{5/2})/KT} + N_{^2D_{5/2}} A_{6731}$$

Aproximando $e^{-E(^4S_{3/2},\,^2D_{5/2})} \sim e^{-E(^4S_{3/2},\,^2D_{3/2})}$

y $e^{-E(^2D_{3/2},\,^2D_{5/2})/KT} \simeq 1$ $h\nu \ll KT$

$$\frac{CN_eN_{^4S_{3/2}}\,\Omega(^4S_{3/2},{}^2D_{5/2})}{g(^4S_{3/2})}\,e^{-E(^4S_{3/2},{}^2D_{5/2})/KT} + \frac{CN_eN_{^2D_{3/2}}\,\Omega(^2D_{3/2},{}^2D_{5/2})}{g(^2D_{3/2})}\,e^{-E(^2D_{3/2},{}^2D_{5/2})/KT} =$$

$$\frac{CN_eN_{^2D_{5/2}}\,\Omega(^2D_{5/2},{}^4S_{3/2})}{g(^2D_{5/2})} + \frac{CN_eN_{^2D_{5/2}}\,\Omega(^2D_{5/2},{}^2D_{3/2})}{g(^2D_{5/2})} + N_{^2D_{5/2}}A_{6716}$$

$\Omega_{ij} = \Omega_{ji}$

(1) ↳
$$\frac{N_{^4S_{3/2}}\,e^{-E(^4S_{3/2},{}^2D_{5/2})/KT}}{g(^4S_{3/2})} = \frac{N_{^2D_{5/2}}\cancel{\Omega(^2D_{5/2},{}^4S_{3/2})}}{g(^2D_{5/2})\cancel{\Omega(^4S_{3/2},{}^2D_{5/2})}} + \frac{N_{^2D_{5/2}}\Omega(^2D_{5/2},{}^2D_{3/2})}{g(^2D_{5/2})\Omega(^4S_{3/2},{}^2D_{5/2})} + \frac{N_{^2D_{5/2}}A_{6716}}{CN_e\Omega(^4S_{3/2},{}^2D_{5/2})} - \frac{N_{^2D_{3/2}}\Omega(^2D_{3/2},{}^2D_{5/2})}{g(^2D_{3/2})\Omega(^4S_{5/2},{}^2D_{5/2})}\cdot 1$$

y

(2)
$$\frac{N_{^4S_{3/2}}\,e^{-E(^4S_{3/2},{}^2D_{3/2})/KT}}{g(^4S_{3/2})} = \frac{N_{^2D_{3/2}}\cancel{\Omega(^2D_{3/2},{}^4S_{3/2})}}{g(^2D_{3/2})\cancel{\Omega(^4S_{3/2},{}^2D_{3/2})}} + \frac{N_{^2D_{3/2}}\Omega(^2D_{3/2},{}^2D_{5/2})}{g(^2D_{3/2})\Omega(^4S_{3/2},{}^2D_{3/2})} + \frac{N_{^2D_{3/2}}A_{6731}}{CN_e\Omega(^4S_{3/2},{}^2D_{3/2})} - \frac{N_{^2D_{5/2}}\Omega(^2D_{3/2},{}^2D_{5/2})}{g(^2D_{5/2})\Omega(^4S_{5/2},{}^2D_{3/2})}$$

$\frac{1}{2} \Rightarrow$

$$1 = \frac{\dfrac{N_{^2D_{5/2}}}{g(^2D_{5/2})} + \dfrac{N_{^2D_{5/2}}\Omega(^2D_{5/2},{}^2D_{3/2})}{g(^2D_{5/2})\Omega(^4S_{3/2},{}^2D_{5/2})} + \dfrac{N_{^2D_{5/2}}A_{6716}}{CN_e\Omega(^4S_{3/2},{}^2D_{5/2})} - \dfrac{N_{^2D_{3/2}}\Omega(^2D_{3/2},{}^2D_{5/2})}{g(^2D_{3/2})\Omega(^4S_{5/2},{}^2D_{5/2})}\cdot 1}{\dfrac{N_{^2D_{3/2}}}{g(^2D_{3/2})} + \dfrac{N_{^2D_{3/2}}\Omega(^2D_{3/2},{}^2D_{5/2})}{g(^2D_{3/2})\Omega(^4S_{3/2},{}^2D_{3/2})} + \dfrac{N_{^2D_{3/2}}A_{6731}}{CN_e\Omega(^4S_{3/2},{}^2D_{3/2})} - \dfrac{N_{^2D_{5/2}}\Omega(^2D_{3/2},{}^2D_{5/2})}{g(^2D_{5/2})\Omega(^4S_{5/2},{}^2D_{3/2})}}$$

$$= \frac{\dfrac{N_{^2D_{5/2}}}{g(^2D_{5/2})}\left\{\dfrac{\Omega(^2D_{5/2},{}^2D_{3/2})}{\Omega(^4S_{3/2},{}^2D_{5/2})} + \dfrac{A_{6716}}{CN_e\Omega(^4S_{3/2},{}^2D_{5/2})}\right\} - \dfrac{N_{^2D_{3/2}}\Omega(^2D_{3/2},{}^2D_{5/2})}{g(^2D_{3/2})\Omega(^4S_{3/2},{}^2D_{5/2})}}{\dfrac{N_{^2D_{3/2}}}{g(^2D_{3/2})}\left\{\dfrac{\Omega(^2D_{3/2},{}^2D_{5/2})}{\Omega(^4S_{3/2},{}^2D_{3/2})} + \dfrac{A_{6731}}{CN_e\Omega(^4S_{3/2},{}^2D_{3/2})}\right\} - \dfrac{N_{^2D_{5/2}}\Omega(^2D_{3/2},{}^2D_{5/2})}{g(^2D_{5/2})\Omega(^4S_{5/2},{}^2D_{3/2})}}$$

$$\frac{N_{^2D_{3/2}}}{g(^2D_{3/2})}\left\{\frac{\Omega(^2D_{3/2},{}^2D_{5/2})}{\Omega(^4S_{3/2},{}^2D_{3/2})} + \frac{A_{6731}}{CN_e\Omega(^4S_{3/2},{}^2D_{3/2})}\right\} + \frac{N_{^2D_{3/2}}\Omega(^2D_{3/2},{}^2D_{5/2})}{g(^2D_{3/2})\Omega(^4S_{3/2},{}^2D_{5/2})}$$

$$= \frac{N_{^2D_{5/2}}}{g(^2D_{5/2})}\left\{\frac{\Omega(^2D_{5/2},{}^2D_{3/2})}{\Omega(^4S_{3/2},{}^2D_{5/2})} + \frac{A_{6716}}{CN_e\Omega(^4S_{3/2},{}^2D_{5/2})}\right\} + \frac{N_{^2D_{5/2}}\Omega(^2D_{3/2},{}^2D_{5/2})}{g(^2D_{5/2})\Omega(^4S_{5/2},{}^2D_{3/2})}$$

$$\frac{N_{^2D_{3/2}}}{g(^2D_{3/2})}\left\{\frac{\Omega(^2D_{3/2},{}^2D_{5/2})}{\Omega(^4S_{3/2},{}^2D_{3/2})} + \frac{A_{6731}}{CN_e\Omega(^4S_{3/2},{}^2D_{3/2})} + \frac{\Omega(^2D_{3/2},{}^2D_{5/2})}{\Omega(^4S_{3/2},{}^2D_{5/2})}\right\}$$

$$= \frac{N_{^2D_{5/2}}}{g(^2D_{5/2})}\left\{\frac{\Omega(^2D_{5/2},{}^2D_{3/2})}{\Omega(^4S_{3/2},{}^2D_{5/2})} + \frac{A_{6716}}{CN_e\Omega(^4S_{3/2},{}^2D_{5/2})} + \frac{\Omega(^2D_{3/2},{}^2D_{5/2})}{\Omega(^4S_{5/2},{}^2D_{3/2})}\right\}$$

$$\frac{N_{^2D_{5/2}}}{N_{^2D_{3/2}}} = \frac{g(^2D_{5/2})}{g(^2D_{3/2})}\,\frac{\left\{\dfrac{\Omega(^2D_{3/2},{}^2D_{5/2})}{\Omega(^4S_{3/2},{}^2D_{3/2})} + \dfrac{A_{6731}}{CN_e\Omega(^4S_{3/2},{}^2D_{3/2})} + \dfrac{\Omega(^2D_{3/2},{}^2D_{5/2})}{\Omega(^4S_{3/2},{}^2D_{5/2})}\right\}}{\left\{\dfrac{\Omega(^2D_{5/2},{}^2D_{3/2})}{\Omega(^4S_{3/2},{}^2D_{5/2})} + \dfrac{A_{6716}}{CN_e\Omega(^4S_{3/2},{}^2D_{5/2})} + \dfrac{\Omega(^2D_{3/2},{}^2D_{5/2})}{\Omega(^4S_{5/2},{}^2D_{3/2})}\right\}}$$

$$\frac{N\,{}^2D_{5/2}}{N\,{}^2D_{3/2}} = \frac{g({}^2D_{5/2})}{g({}^2D_{3/2})}\left(\frac{\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},{}^2D_{3/2})}+\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},{}^2D_{5/2})}}{\dfrac{\Omega({}^2D_{5/2},{}^2D_{3/2})}{\Omega({}^4S_{3/2},{}^2D_{5/2})}+\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},{}^2D_{3/2})}}\right) + \frac{g({}^2D_{5/2})}{g({}^2D_{3/2})}\,\frac{\dfrac{A_{6731}}{\cancel{CN_e}\,\Omega({}^4S_{3/2},{}^2D_{3/2})}}{\dfrac{A_{6716}}{\cancel{CN_e}\,\Omega({}^4S_{3/2},{}^2D_{5/2})}}$$

$$\frac{N\,{}^2D_{5/2}}{N\,{}^2D_{3/2}} = \frac{g({}^2D_{5/2})}{g({}^2D_{3/2})}\left(1+\frac{\Omega({}^4S_{3/2},{}^2D_{5/2})\,A_{6731}}{\Omega({}^4S_{3/2},{}^2D_{3/2})\,A_{6716}}\right)$$

coeficientes de emisión expontanea

$$\frac{N\,{}^2D_{5/2}}{N\,{}^2D_{3/2}} = \frac{g({}^2D_{5/2})}{g({}^2D_{3/2})}\,\frac{\left\{1+\dfrac{A_{6731}\,g({}^2D_{3/2})}{CN_e\,\Omega({}^4S_{3/2},{}^2D_{3/2})}+\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},{}^2D_{3/2})}+\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},{}^2D_{5/2})}\right\}}{\left\{1+\dfrac{A_{6716}\,g({}^2D_{5/2})}{CN_e\,\Omega({}^4S_{3/2},{}^2D_{5/2})}+\dfrac{\Omega({}^2D_{5/2},{}^2D_{3/2})}{\Omega({}^4S_{3/2},{}^2D_{5/2})}+\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},{}^2D_{3/2})}\right\}}$$

Finalmente:

- $$\frac{J_{6716}}{J_{6731}} = \frac{A_{6716}\,g({}^2D_{5/2})}{A_{6731}\,g({}^2D_{3/2})}\,\frac{\left\{1+\dfrac{A_{6731}\,g({}^2D_{3/2})}{CN_e\,\Omega({}^4S_{3/2},{}^2D_{3/2})}+\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},{}^2D_{3/2})}+\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},{}^2D_{5/2})}\right\}}{\left\{1+\dfrac{\Omega({}^2D_{5/2},{}^2D_{3/2})}{\Omega({}^4S_{3/2},{}^2D_{5/2})}+\dfrac{A_{6716}\,g({}^2D_{5/2})}{CN_e\,\Omega({}^4S_{3/2},{}^2D_{5/2})}+\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},D_{3/2})}\right\}}$$

y Para OII

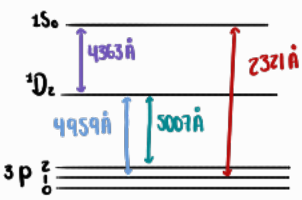
- $$\frac{J_{3729}}{J_{3726}} = \frac{A_{3729}\,g({}^2D_{5/2})}{A_{3726}\,g({}^2D_{3/2})}\,\frac{\left\{1+\dfrac{A_{3726}\,g({}^2D_{3/2})}{CN_e\,\Omega({}^4S_{3/2},{}^2D_{3/2})}+\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},{}^2D_{3/2})}+\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},{}^2D_{5/2})}\right\}}{\left\{1+\dfrac{\Omega({}^2D_{5/2},{}^2D_{3/2})}{\Omega({}^4S_{3/2},{}^2D_{5/2})}+\dfrac{A_{3729}\,g({}^2D_{5/2})}{CN_e\,\Omega({}^4S_{3/2},{}^2D_{5/2})}+\dfrac{\Omega({}^2D_{3/2},{}^2D_{5/2})}{\Omega({}^4S_{3/2},D_{3/2})}\right\}}$$

Estimación de la Temperatura electrónica

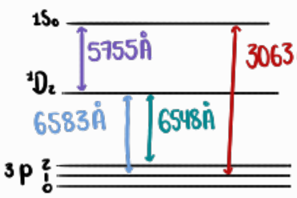
[OIII]

multiplicidad = 2S+1

L_J



[NII]



Buscamos las ecuaciones que describen las transiciones de estos dos iones

Nivel S: $q_{P\to S} + q_{D\to S} = q^e_{S\to D} + q^e_{S\to P} + q_{S\to D} + q_{S\to P}$

Nivel D: $q_{P\to D} + q_{S\to D} + q^e_{S\to D} = q_{D\to S} + q_{D\to P} + q^e_{D\to P}$

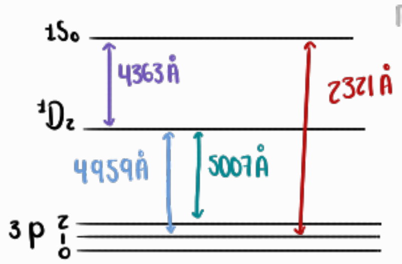
$$\vec{q}_{kj} = N_{ik} A_{kj}$$

$$q_{kj} = \frac{N_{ik} N_e}{T^{1/2}} \frac{8.6\times10^{-6}\,\Omega_{jk}}{g_k}$$

$$q_{jk} = N_{ij} N_e \frac{8.6\times10^{-6}\,\Omega_{jk}}{T^{1/2}\; g_j} e^{-E_{jk}/KT}$$

$$C = \frac{8.63\times10^{-6}}{T^{1/2}}$$

[OIII]



Necesitamos ponerlo en términos de los observables ⇒ los fotones

Nivel S =

$$q_{P\to S} + q_{D\to S} = q^e_{S\to D} + q^e_{S\to P} + q_{S\to D} + q_{S\to P}$$

$$N_P C N_e \frac{\Omega_{PS}}{g_P} e^{-E_{PS}/KT} + N_D C N_e \frac{\Omega_{DS}}{g_D} e^{-E_{DS}/KT} = N_S A_{SD} + N_S A_{SP} + N_S C N_e \frac{\Omega_{SD}}{g_S} + N_S C N_e \frac{\Omega_{SP}}{g_S} \qquad (A)$$

Nivel D (4.1.51)

$$N_P C N_e \frac{\Omega_{PD}}{g_P} e^{-E_{PD}/KT} + N_S C N_e \frac{\Omega_{SD}}{g_S} + N_S A_{SD} = N_D C N_e \frac{\Omega_{DS}}{g_D} e^{-E_{DS}/KT} + N_D C N_e \frac{\Omega_{DP}}{g_D} + N_D A_{DP} \qquad (B)$$

Teniendo en cuenta que $E_{PS} = E_{PD} + E_{DS}$, Buscamos $\frac{N_D}{N_S}$

(A):

$$N_D C N_e \frac{\Omega_{DS}}{g_D} e^{-E_{DS}/KT} = N_S A_{SD} + N_S A_{SP} + N_S C N_e \frac{\Omega_{SD}}{g_S} + N_S C N_e \frac{\Omega_{SP}}{g_S} - N_P C N_e \frac{\Omega_{PS}}{g_P} e^{-(E_{PD}+E_{DS})/KT}$$

(B)

$$N_D C N_e \frac{\Omega_{DS}}{g_D} e^{-E_{DS}/KT} = N_P C N_e \frac{\Omega_{PD}}{g_P} e^{-E_{PD}/KT} + N_S C N_e \frac{\Omega_{SD}}{g_S} + N_S A_{SD} - N_D C N_e \frac{\Omega_{DP}}{g_D} - N_D A_{DP}$$

$\frac{A}{B}$

$$N_S A_{SD} + N_S A_{SP} + N_S C N_e \frac{\Omega_{SD}}{g_S} + N_S C N_e \frac{\Omega_{SP}}{g_S} - N_P C N_e \frac{\Omega_{PS}}{g_P} e^{-(E_{PD}+E_{DS})/KT} =$$

$$N_P C N_e \frac{\Omega_{PD}}{g_P} e^{-E_{PD}/KT} + N_S C N_e \frac{\Omega_{SD}}{g_S} + N_S A_{SD} - N_D C N_e \frac{\Omega_{DP}}{g_D} - N_D A_{DP}$$

$$\frac{N_D}{N_S}=\frac{\frac{(A_{SD}+A_{SP})}{N_e}\frac{\Omega_{PD}}{\Omega_{PS}}e^{E_{SD}/kT}+\frac{C}{g_S}(\Omega_{SD}+\Omega_{SP})\frac{\Omega_{PD}}{\Omega_{PS}}e^{E_{SD}/kT}+\frac{C\Omega_{SD}}{g_S}+\frac{A_{SD}}{N_e}}{\frac{C\Omega_{DS}}{g_D}e^{-E_{DS}/kT}+\frac{C\Omega_{DP}}{g_D}+\frac{A_{DP}}{N_e}+\frac{C\Omega_{DS}\Omega_{PD}}{g_D\Omega_{PS}}}$$

Multiplicando por $\dfrac{e^{-E_{SD}/kT}g_S}{C\Omega_{PD}}$

$$\left(\frac{(A_{SD}+A_{SP})}{N_e}\frac{\Omega_{PD}}{\Omega_{PS}}e^{E_{SD}/kT}+\frac{C}{g_S}(\Omega_{SD}+\Omega_{SP})\frac{\Omega_{PD}}{\Omega_{PS}}e^{E_{SD}/kT}+\frac{C\Omega_{SD}}{g_S}+\frac{A_{SD}}{N_e}\right)\frac{e^{-E_{SD}/kT}g_S}{C\Omega_{PD}}$$

$$\hookrightarrow=\frac{(A_{SD}+A_{SP})g_S}{N_e\Omega_{PS}}+\frac{C(\Omega_{SD}+\Omega_{SP})}{\Omega_{PS}}+\frac{\Omega_{SD}}{\Omega_{PD}}e^{-E_{SD}/kT}+\frac{A_{SD}}{N_e}\frac{e^{-E_{SD}/kT}g_S}{C\Omega_{PD}}\Bigg)$$

$$\left(\frac{C\Omega_{DS}}{g_D}e^{-E_{DS}/kT}+\frac{C\Omega_{DP}}{g_D}+\frac{A_{DP}}{N_e}+\frac{C\Omega_{DS}\Omega_{PD}}{g_D\Omega_{PS}}\right)\frac{e^{-E_{SD}/kT}g_S}{C\Omega_{PD}}$$

$$\hookrightarrow\frac{\Omega_{DS}}{g_D}e^{-E_{DS}/kT}e^{-E_{SD}/kT}g_S+\frac{g_S}{g_D}e^{-E_{SD}/kT}$$

$$\boxed{\frac{N_D}{N_S}=\frac{\frac{(A_{SD}+A_{SP})g_S}{CN_e\Omega_{PS}}+\frac{\Omega_{SD}}{\Omega_{PD}}e^{-E_{SD}/kT}+\frac{A_{SD}g_S}{CN_e\Omega_{PD}}e^{-E_{SD}/kT}+\frac{\Omega_{SD}+\Omega_{SP}}{\Omega_{PS}}}{\frac{g_S}{g_D}e^{-E_{SD}/kT}\left[\frac{\Omega_{DS}}{\Omega_{PD}}e^{-E_{DS}/kT}+\frac{A_{DP}g_D}{CN_e\Omega_{PD}}+\frac{\Omega_{DS}}{\Omega_{PS}}+1\right]}}$$

$$\rightarrow \frac{\Omega_{DS}}{\Omega_{PS}} = \gamma - 1$$

Si $\gamma = \dfrac{\Omega_{DS} + \Omega_{PS}}{\Omega_{PS}} = \dfrac{\Omega_{DS}}{\Omega_{PS}} + 1$

$$\frac{N_D}{N_S} = \frac{\dfrac{(A_{SD} + A_{SP})\, g_S}{C N_e \Omega_{PS}} + \dfrac{\Omega_{SD}}{\Omega_{PD}} e^{-E_{SD}/kT} + \dfrac{A_{SD}\, g_S}{C N_e \Omega_{PD}} e^{-E_{SD}/kT} + \overset{\gamma}{\dfrac{\Omega_{SD} + \Omega_{SP}}{\Omega_{PS}}}}{\dfrac{g_S}{g_D} e^{-E_{SD}/kT} \left[\dfrac{\Omega_{DS}}{\Omega_{PD}} e^{-E_{DS}/kT} + \dfrac{A_{DP}\, g_D}{C N_e \Omega_{PD}} + \underbrace{\dfrac{\Omega_{DS}}{\Omega_{PS}}}_{\gamma - 1} + 1 \right]}$$

$$= \frac{\cancel{\gamma} \left[\dfrac{(A_{SD} + A_{SP})\, g_S}{\gamma C N_e \Omega_{PS}} + \dfrac{\Omega_{SD}}{\gamma \Omega_{PD}} e^{-E_{SD}/kT} + \dfrac{A_{SD}\, g_S}{\gamma C N_e \Omega_{PD}} e^{-E_{SD}/kT} + 1 \right]}{\cancel{\gamma} \; \dfrac{g_S}{g_D} e^{-E_{SD}/kT} \left[\dfrac{\Omega_{DS}}{\Omega_{PD}\gamma} e^{-E_{DS}/kT} + \dfrac{A_{DP}\, g_D}{\gamma C N_e \Omega_{PD}} + \gamma \right]}$$

$$\frac{J_1}{J_2} = \frac{A_1 \left[\dfrac{(A_{SD} + A_{SP})\, g_S}{\gamma C N_e \Omega_{PS}} + \dfrac{\Omega_{SD}}{\gamma \Omega_{PD}} e^{-E_{SD}/kT} + \dfrac{A_{SD}\, g_S}{\gamma C N_e \Omega_{PD}} e^{-E_{SD}/kT} + 1 \right]}{A_2 \; \dfrac{g_S}{g_D} e^{-E_{SD}/kT} \left[\dfrac{\Omega_{DS}}{\Omega_{PD}\gamma} e^{-E_{DS}/kT} + \dfrac{A_{DP}\, g_D}{\gamma C N_e \Omega_{PD}} + \gamma \right]}$$

$$\frac{D_N}{D_S} = \frac{\left[\dfrac{(A_{SD} + A_{SP})\, g_S}{\gamma C N_e \Omega_{PS}} + \dfrac{\Omega_{SD}}{\gamma \Omega_{PD}} e^{-E_{SD}/kT} + \dfrac{A_{SD}\, g_S}{\gamma C N_e \Omega_{PD}} e^{-E_{SD}/kT} + 1 \right]}{\dfrac{g_S}{g_D} e^{-E_{SD}/kT} \left[\dfrac{\Omega_{DS}}{\Omega_{PD}\gamma} e^{-E_{DS}/kT} + \dfrac{A_{DP}\, g_D}{\gamma C N_e \Omega_{PD}} + \gamma \right]}$$

Multiplicando por $\dfrac{A_{DP} h\nu_{DP}}{A_{SD} h\nu_{SD}}$

$$\frac{j_{DP}}{j_{SD}} = \frac{\left[\dfrac{(A_{SD} + A_{SP})\, g_S}{\gamma C N_e \Omega_{PS}} + \dfrac{\Omega_{SD}}{\gamma \Omega_{PD}} e^{-E_{SD}/kT} + \dfrac{A_{SD}\, g_S}{\gamma C N_e \Omega_{PD}} e^{-E_{SD}/kT} + 1 \right]}{\dfrac{g_S}{g_D} e^{-E_{SD}/kT} \left[\dfrac{\Omega_{DS}}{\Omega_{PD}\gamma} e^{-E_{DS}/kT} + \dfrac{A_{DP}\, g_D}{\gamma C N_e \Omega_{PD}} + \gamma \right]} \; \frac{A_{PD}\, h \nu_{DP}}{A_{SD}\, h \nu_{SD}}$$

Para los iones que estamos considerando:

$$\frac{J_{6716}}{J_{6731}} = \frac{F_{6716}}{F_{6731}} = \frac{3}{2}\,\frac{A_{6716}}{A_{6731}}\left(\frac{N_eC + 0.26A_{6731}}{N_eC + 0.26A_{6716}}\right)$$

$$\frac{J_{3729}}{J_{3726}} = \frac{F_{3729}}{F_{3726}} = \frac{3}{2}\,\frac{A_{3729}}{A_{3726}}\left(\frac{N_eC + 1.6\,A_{3729}}{N_eC + 1.6\,A_{3726}}\right)$$

$$\frac{F_{5007} + F_{4959}}{F_{4363}} = 0.054\,e^{\frac{32.985}{T}}\;\frac{\frac{N_e}{\sqrt{T}} + 2.6\times10^{5}\left(1+0.12\,e^{-\frac{32.985}{T}}\right)}{\frac{N_e}{\sqrt{T}} + 1.627}$$

$$\frac{F_{6543} + F_{6583}}{F_{5755}} = 0.01\,e^{\frac{25.007}{T}}\;\frac{\frac{N_e}{\sqrt{T}} + 1.3\times10^{5}\left(1+0.12\,e^{-\frac{25.007}{T}}\right)}{\frac{N_e}{\sqrt{T}} + 250.4}$$

Tabla 1. Coeficientes de Einstein para las transiciones:

Transiciones:	Especie	$\lambda(\AA)$	$A(s^{-1})$	Especie	$\lambda(\AA)$	$A(s^{-1})$
$^1D_2$ $^1S_0$	OIII	4363	1.6	NII	5755	1
$^3P_2$ $^1D_2$	OIII	5007	2×10^{-2}	NII	6583	3×10^{-3}
$^3P_1$ $^1D_2$	OIII	4959	6.8×10^{-3}	NII	6548	9.8×10^{-4}
$^3P_1$ $^1S_0$	OIII	2321	2.3×10^{-1}	NII	3063	3.3×10^{-2}
$^4S_{3/2}$ $^2D_{5/2}$	SII	6716	2.6×10^{-4}	OII	3729	3.6×10^{-5}
$^4S_{3/2}$ $^2D_{3/2}$	SII	6731	8.8×10^{-4}	OII	3726	1.6×10^{-4}